5.1 intro to factor market

DDDDD
CCCAC
ADBDB
ACCCA
A

1. Assume the market for disposable coffee cups is in equilibrium and disposable coffee cups are inputs for serving brewed coffee. Which of the following will result in a higher short-run equilibrium price of disposable coffee cups?

(A) A decrease in the supply of coffee
(B) A decrease in the number of locations serving brewed coffee
(C) An increase in the supply of disposable coffee cups
(D) An increase in the demand for brewed coffee ✓
(E) An increase in the price of tea, a complement for coffee

Answer D

Correct. An increase in the demand for brewed coffee will shift the demand curve for disposable coffee cups to the right, resulting in a higher equilibrium price and equilibrium quantity of disposable coffee cups.

2. A factor of production will NOT earn economic rent when its supply is

(A) elastic
(B) inelastic
(C) unit elastic
(D) perfectly elastic ✓
(E) perfectly inelastic

3. A firm's demand curve for labor is equal to a segment of its

(A) average variable cost curve
(B) total revenue curve
(C) marginal cost curve
(D) marginal revenue product curve ✓
(E) average product curve

4. A firm's demand for labor is known as a derived demand because

(A) the firm gains utility from hiring more labor
(B) the wage rate paid to workers depends on the demand for labor
(C) the amount of labor demanded depends on the amount of capital invested
(D) the amount of labor demanded depends on the demand for the firm's product ✓
(E) the firm will benefit from hiring additional labor

5.1 intro to factor market

5. Assume that barber shops operate in perfectly competitive product and factor markets. Which of the following will happen to working barbers if the price of haircuts decreases?

(A) The marginal product of the last barber hired will not change.

(B) The marginal product of the last barber hired will decrease.

(C) The marginal revenue product curve will shift to the right.

(D) The marginal revenue product curve will shift to the left. ✓

(E) There will be a movement upward along the marginal revenue curve.

6. Assume that firms sell their output in a perfectly competitive product market and hire labor in a perfectly competitive labor market. If all other factors remain constant, an increase in the demand for the firms' product will result in which of the following changes in the labor market?

(A) Firms will move down along the demand curve for labor and hire more workers.

(B) The supply curve for labor will shift to the right.

(C) The demand curve for labor will shift to the right. ✓

(D) The supply curve for labor will shift to the left.

(E) The demand curve for labor will shift to the left.

7.

Number of Workers	Total Product
1	15
2	20
3	24
4	27
5	29

The table above shows the short-run output for a perfectly competitive firm. If the price of the product is $10, what is the marginal revenue product of the third worker hired?

(A) $24

(B) $27

(C) $40 ✓

(D) $240

(E) $300

8. Which of the following illustrates how factor demand is related to product demand?

5.1 intro to factor market

(A) The quantity demanded for restaurant meals depends on the price of the meal.
(B) The demand for books depends on the number of authors in the market.
(C) The demand for airline pilots depends on the demand for air travel. ✓
(D) The demand for auto mechanics depends on the supply of auto parts.
(E) The demand for health care depends on the cost of health care.

Answer C

Correct. Pilots can be considered a factor of production in the airline industry. When demand for a product or service (such as air travel) increases, it can lead to an increase in the demand for factors (such as pilots) used to produce that product or service.

9. When the demand for new homes decreases, the demand for construction workers who build homes decreases. This relationship illustrates the concept of

(A) derived demand ✓
(B) diminishing marginal productivity of labor
(C) substitution in production
(D) supply shock
(E) property rights

10. Economists argue that most professional athletes earn economic rent because they

(A) make additional income through commercial endorsements of products
(B) are able to participate in sports for only a limited number of years before changing occupations
(C) earn far more as professional athletes than they could earn in their next-best occupation ✓
(D) participate in sporting events only about six months during a year
(E) work less than 40 hours a week during the sport season

11. For a firm hiring labor in a perfectly competitive labor market, the marginal revenue product curve slopes downward after some point because as more of a factor is employed, which of the following declines?

(A) Marginal product ✓
(B) Marginal factor cost
(C) Marginal cost
(D) Total output
(E) Wage rates

5.1 intro to factor market

12. Heggood, Inc., a perfectly competitive firm, produces hats using capital and labor purchased in competitive factor markets. If the demand for hats increases, Heggood, Inc. will

(A) decrease the price of hats

(B) decrease its demand for labor

(C) decrease the quantity of labor demanded

(D) increase its demand for capital ✓

(E) increase the rental price of capital

Answer D

Correct. Since the demand for capital used to produce hats is derived from the demand for hats, an increase in the demand for hats will cause an increase in the demand for capital used to produce hats.

13. A firm's decision to hire a factor of production DOES NOT depend on which of the following?

(A) The price of the product produced by the factor input

(B) The average product of the factor input ✓

(C) The price of the factor input

(D) The demand for the product the factor produces

(E) The marginal product of the factor input

Answer B

Correct. Neither a firm's decision to hire a factor nor its demand for a factor depends on the average product of the factor. A firm's decision to hire a factor of production depends on the productivity of the factor, as measured by the marginal product of the factor, the factor's price (or marginal factor cost), and the price of the firm's product (or the demand for the product the factor produces).

14. If labor is the only variable input of a firm and the marginal product of labor is falling, the firm will always produce

(A) more than the profit-maximizing level of output

(B) less than the profit-maximizing level of output

(C) at a level of output where average total cost is at a minimum

(D) at a level of output where marginal costs are rising ✓

(E) at a level of output where average variable costs are falling

5.1 intro to factor market

Question is based on the chart below for a firm that is perfectly competitive in both the labor and product markets, showing how much daily output a firm can produce using different numbers of workers.

Number of Workers	Output
1	3
2	9
3	16
4	21
5	23
6	24

15. If output sells for $20 per unit, what is the marginal revenue product of the fifth worker?

(A) $2

(B) $40 ✓

(C) $100

(D) $115

(E) $460

16. Which of the following statements about the relationship between the demand for apple pickers and the demand for apples is true?

(A) An increase in the demand for apples increases the demand for apple pickers. ✓

(B) An increase in the demand for apples increases the wage rate and decreases the demand for apple pickers.

(C) A decrease in the demand for apples decreases the wage rate and increases the demand for apple pickers.

(D) An increase in the demand for apples increases productivity and the demand for apple pickers.

(E) A decrease in the demand for apples decreases productivity and the demand for apple pickers.

Answer A

Correct. The demand for a factor of production such as labor depends on the demand for the product that labor is used to produce. Thus, an increase in the demand for apples will increase the demand for apple

5.1 intro to factor market

pickers.

17. The demand curve for labor shows which of the following?

(A) The quantity of output each worker can produce at various wage rates
(B) The number of workers required to produce a given level of output
(C) The number of workers a firm is willing and able to hire at various wage rates ✓
(D) The positive relationship between the wage rate and the number of hours people wish to work
(E) The number of workers who are willing and able to work at various wage rates

Answer C

Correct. The demand curve for labor is downward sloping, and it shows the number of workers firms (employers) are willing to hire at each possible wage rate. The quantity of labor demanded is negatively related to the wage rate.

18. Which of the following will occur when wage rates decrease in a given labor market?

(A) The supply of labor will decrease.
(B) The demand for labor will increase.
(C) The quantity supplied of labor will decrease. ✓
(D) The quantity demanded of labor will decrease.
(E) The supply of labor will decrease and the demand for labor will increase.

Answer C

Correct. There is a positive relationship between the wage rate and the quantity supplied of labor. A decrease in the wage rate will decrease the quantity supplied of labor, moving down along the labor supply curve.

19. Which of the following will occur in a given labor market when the wage rate rises?

5.1 intro to factor market

(A) The quantity demanded of labor will increase.
(B) The demand for labor will decrease.
(C) The quantity of labor supplied will increase. ✓
(D) The supply of labor will decrease.
(E) The supply of labor will increase.

Answer C

Correct. An increase in the wage rate causes a movement up along a given labor supply curve, resulting in an increase in the quantity of labor supplied. There is a positive relationship between the wage rate and the quantity of labor supplied.

The following questions are based on the following table below, which shows the relationship between the number of workers and coal output (in tons per day).

Number of Workers	Total Output of Coal
0	0
1	25
2	44
3	60
4	70
5	75

20. The marginal physical product of the second worker is

(A) 19 ✓
(B) 22
(C) 25
(D) 44
(E) 75

---

21. The concept of derived demand is described by which of the following?

5.1 intro to factor market

(A) A decrease in the demand for theater tickets will decrease the demand for actresses and actors. ✓

(B) If the salaries of basketball players increase, the quantity of basketball players demanded will decrease.

(C) An increase in the income of consumers will increase the demand for opera tickets.

(D) An increase in the demand for movie tickets will decrease the demand for video rentals.

(E) A decrease in the price of movie tickets will increase the demand for movie tickets.